

```

#include <stdio.h>
#include <stdint.h>
#include <string.h>

// ===== Hex-Segment 정의 =====
const char* HEX_SEGMENTS[] = {
    "Idle", "Conv2D", "MatMul", "ReLU", "Softmax", "BatchNorm", "DepthwiseConv", "GRU",
    "Attention"
};

// ===== RV-SVE ISA 정의 =====
#define OPCODE_RV_SVE 0b1011011
#define FUNCT3_VMUL_VV 0b010
#define FUNCT7_VECTOR 0b0000000

// ===== RV-SVE 명령어 구조 =====
typedef struct {
    uint64_t raw_inst;
    int rd, rs1, rs2;
    int funct3;
    int is_vector;
    char opcode[16];
} rv_sve_uop;

// ===== NPU 컨텍스트 구조 =====
typedef struct {
    rv_sve_uop uop;
    const char* segmentCode;
    const char* operation;
    int tensorH, tensorW;
    int loadEstimate;
    char priority[8];
    char token[32];
} NPUContext;

// ===== 명령어 디코딩 + 연산 목적 분석 =====
NPUContext decode_segment(uint64_t raw, const char* hexCode, int h, int w) {
    NPUContext ctx;
    ctx.segmentCode = hexCode;
    ctx.tensorH = h;
    ctx.tensorW = w;
    ctx.loadEstimate = h * w;

    int segID = strtol(hexCode, NULL, 16);
    ctx.operation = HEX_SEGMENTS[segID];

    ctx.uop.raw_inst = raw;
    ctx.uop.rd = (raw >> 7) & 0x1F;

```

```

    ctx.uop.rs1 = (raw >> 15) & 0x1F;
    ctx.uop.rs2 = (raw >> 20) & 0x1F;
    ctx.uop.funct3 = (raw >> 12) & 0x7;
    ctx.uop.is_vector = ((raw & 0x7F) == OPCODE_RV_SVE);
    strcpy(ctx.uop.opcode, (ctx.uop.is_vector && ctx.uop.funct3 == FUNCT3_VMUL_VV) ?
"VMUL.VV" : "NOP");

    return ctx;
}

// ===== 중요도 판단 =====
NPUContext rank_importance(NPUContext ctx) {
    strcpy(ctx.priority, ctx.loadEstimate > 4096 ? "high" : "mid");
    return ctx;
}

// ===== 디스패치 정책 =====
NPUContext dispatch_tensor_core(NPUContext ctx) {
    sprintf(ctx.token, "EXEC_%s_%s", ctx.operation, strcmp(ctx.priority, "high") == 0 ? "FAST"
: "IDLE");
    return ctx;
}

// ===== 실행 흐름 추적 =====
void trace_npu_cycle(NPUContext ctx) {
    printf("[NPU] %s dispatched for %s (%dx%d tensor)\n", ctx.token, ctx.operation,
ctx.tensorH, ctx.tensorW);
    printf("[RV-SVE] %s r%d ← r%d, r%d\n", ctx.uop.opcode, ctx.uop.rd, ctx.uop.rs1,
ctx.uop.rs2);
}

// ===== NPU 파이프라인 IPC =====
void run_npu_pipeline(uint64_t raw_inst, const char* hexCode, int h, int w) {
    NPUContext ctx = decode_segment(raw_inst, hexCode, h, w);
    ctx = rank_importance(ctx);
    ctx = dispatch_tensor_core(ctx);
    trace_npu_cycle(ctx);
}

// ===== 테스트 =====
int main() {
    // VMUL.VV 명령어 생성
    uint64_t raw = ((uint64_t)FUNCT7_VECTOR << 25) | ((uint64_t)2 << 20) |
        ((uint64_t)1 << 15) | ((uint64_t)FUNCT3_VMUL_VV << 12) |
        ((uint64_t)0 << 7) | ((uint64_t)OPCODE_RV_SVE);

    run_npu_pipeline(raw, "0x02", 128, 128); // MatMul
    return 0;
}

```

